# 15B 1 CARBONYL COMPOUNDS AND THEIR PHYSICAL PROPERTIES

SPECIFICATION REFERENCE
15.6 15.7(i) 15.7(ii)

## LEARNING OBJECTIVES

- Understand the nomenclature of aldehydes and ketones and be able to draw their structural, displayed and skeletal formulae.
- Understand that aldehydes and ketones:
  (i) do not form intermolecular hydrogen bonds, and this affects their physical properties
  (ii) can form hydrogen bonds with water, and this affects their solubility.

### LEARNING TIP

When naming ketones with a chain of five or more carbon atoms, a locant is needed to show the position of the carbonyl group. When naming aldehydes, a locant is never used to show the position of the carbonyl group because the CHO group must be at the end of the chain.

### DID YOU KNOW?

Some naturally occurring aldehydes and ketones are important in living systems. Some function as sex hormones; others are used as flavourings. For example, testosterone, the male sex hormone, and progesterone, a hormone secreted by the ovaries that stimulates the growth of cells in the uterine wall, both contain a ketone functional group. Aldehydes can be found in foods, perfumes and medicines. Some aldehydes are responsible for very pleasant tastes and odours, such as vanillin from vanilla beans, benzaldehyde from almonds and cinnamaldehyde from cinnamon. About 82% of the chemicals present in coriander leaves are aldehydes with 9 or 10 carbon atoms in the molecules.

## NOMENCLATURE

We met carbonyl compounds briefly in **Book 1**, but only as the products of the oxidation of alcohols.

The carbonyl group (C=O) is a carbon atom joined by a double bond to an oxygen atom. Many types of compounds contain this group (such as carboxylic acids, which contain COOH), but only two types are classified as carbonyl compounds – these are aldehydes and ketones.

### ALDEHYDES

A carbonyl compound is described as an aldehyde if there is a hydrogen atom joined to the carbonyl group. The naming system uses the suffix *-al*, and the carbon chain is named in the same way as an alkane. If locants are used, then the carbon of the carbonyl group is carbon 1 in the chain. **Table A** shows some examples.

| STRUCTURAL FORMULA | NAME | COMMON NAME |
|---|---|---|
| HCHO | methanal | formaldehyde |
| $CH_3CHO$ | ethanal | acetaldehyde |
| $CH_3CH_2CHO$ | propanal | (propionaldehyde) |
| $(CH_3)_2CHCHO$ | methylpropanal | (isobutyraldehyde) |
| $CH_3CH_2CH_2CHO$ | butanal | (butyraldehyde) |
| $C_6H_5CHO$ | benzenecarbaldehyde | benzaldehyde |

**table A**

The general formula is RCHO.

### KETONES

A carbonyl compound is described as a ketone if there are only hydrocarbon groups joined to the carbonyl group. The naming system uses the suffix *-one*, and the carbon chain is named in the same way as an alkane. If locants are used, then one of the end carbon atoms in the chain of the carbonyl group is carbon 1 in the chain. The rules about using the lowest possible numbers, as with alkanes, apply to ketones. **Table B** shows some examples.

| STRUCTURAL FORMULA | NAME | COMMON NAME |
|---|---|---|
| $CH_3COCH_3$ | propanone | acetone |
| $CH_3COCH_2CH_3$ | butanone | (methyl ethyl ketone) |
| $CH_3COCH_2CH_2CH_3$ | pentan-2-one | (methyl propyl ketone) |
| $CH_3CH_2COCH_2CH_3$ | pentan-3-one | (diethyl ketone) |
| $(CH_3)_2CHCOCH_3$ | 3-methylbutan-2-one | (methyl isopropyl ketone) |

**table B**

The general formula is RCOR′.

For both aldehydes and ketones, the common names in brackets are rarely used.

## DISPLAYED AND SKELETAL FORMULAE OF ALDEHYDES AND KETONES

**Table C** and **table D** show the displayed and skeletal formulae for some aldehydes and ketones.

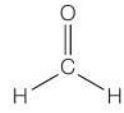
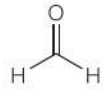
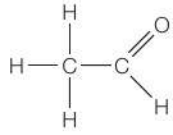
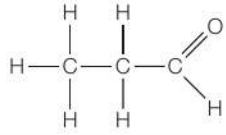
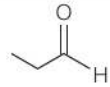
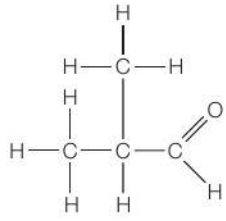
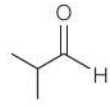
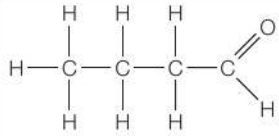
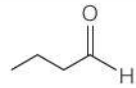
| NAME | DISPLAYED FORMULA | SKELETAL FORMULA |
|---|---|---|
| methanal |  |  |
| ethanal |  |  |
| propanal |  |  |
| methylpropanal |  |  |
| butanal |  |  |

**table C**

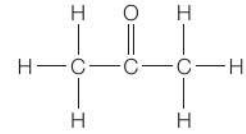
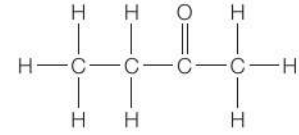
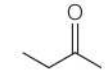
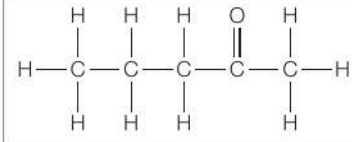
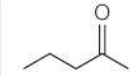
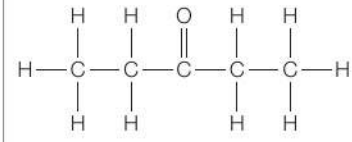
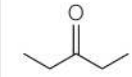
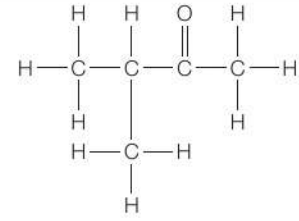
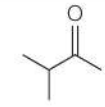
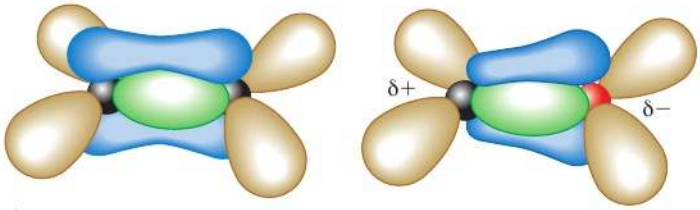
| NAME | DISPLAYED FORMULA | SKELETAL FORMULA |
|---|---|---|
| propanone |  |  |
| butanone |  |  |
| pentan-2-one |  |  |
| pentan-3-one |  |  |
| 3-methylbutan-2-one |  |  |

**table D**

## BONDING

You are familiar with the nature of the C=C double bond in alkenes. It is composed of a combination of sigma and pi bonding. However, the bond is non-polar because there are two identical (carbon) atoms involved. The situation in the carbonyl group is very similar, except that the bond is polar because of the differing electronegativities of carbon and oxygen. **Fig A** compares the two.




▲ **fig A** The bonding in an alkene is on the left. The bonding in a carbonyl compound is on the right. Green is used to represent sigma bonding and blue is used to represent pi bonding.

Note that the electron density in the pi bond of the alkene is evenly distributed across both carbon atoms, but that in the carbonyl compound the electron density is greater near the oxygen atom than near the carbon atom.

## PHYSICAL PROPERTIES

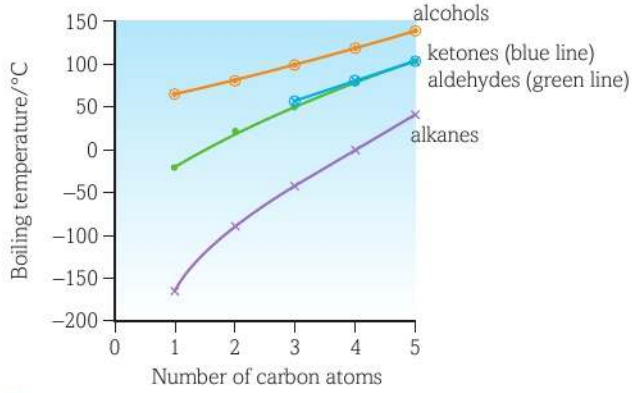
Aldehydes have distinctive smells. Those with short carbon chains have unpleasant smells (butanal smells of rancid butter). Some of those with long carbon chains have very pleasant smells and are used in expensive perfumes (nonanal, with nine carbon atoms per molecule, smells of roses). The smells of the lower ketones, especially propanone, remind most people of solvents. In fact, propanone is widely used as a solvent, including as nail varnish remover.



▲ **fig B** Cinnamon (which contains the alehyde cinnamaldehyde) comes from the dried bark of the tree *Cinnamomum zeylanicum*. It is widely used as a spice in cookery.

## BOILING TEMPERATURES

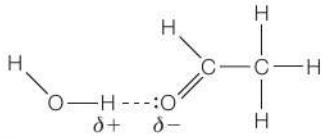
Aldehydes and ketones contain the polar C=O group and so have permanent dipole–dipole interactions, as well as London forces. They do not have hydrogen bonding because all of their hydrogen atoms are joined to carbon atoms. As with other homologous series, for example, alkanes and alcohols, boiling temperatures increase with increasing molar mass as the extent of London forces increases. At room temperature, methanal is a gas. The other carbonyl compounds are liquids. **Fig C** compares the boiling temperatures of some carbonyl compounds.




▲ **fig C** The boiling temperatures of some alkanes, aldehydes, ketones and alcohols. The effect of the intermolecular forces on the boiling temperatures can be seen clearly, with dipole–dipole attractions, which result from the carbonyl groups of the aldehydes and ketones, causing higher boiling temperatures than the equivalent alkanes but lower than the alcohols with their hydrogen bonding.

## SOLUBILITY IN WATER

The lower (shorter chain) aldehydes and ketones are soluble in water because they can form hydrogen bonds (shown as a dashed line) with water molecules. **Fig D** shows hydrogen bonding between ethanal and water.




▲ **fig D** Hydrogen bonding between water molecules and ethanal molecules is stronger than the intermolecular bonds in the pure aldehyde.

The solubility of aldehydes and ketones decreases with increasing chain length, as the hydrocarbon part of the molecules becomes more significant.

### CHECKPOINT

1. Write the displayed formula for:
   (a) propanal
   (b) propanone.
2. Draw a diagram to explain why propanone is soluble in water.

# 15B 2 REDOX REACTIONS OF CARBONYL COMPOUNDS

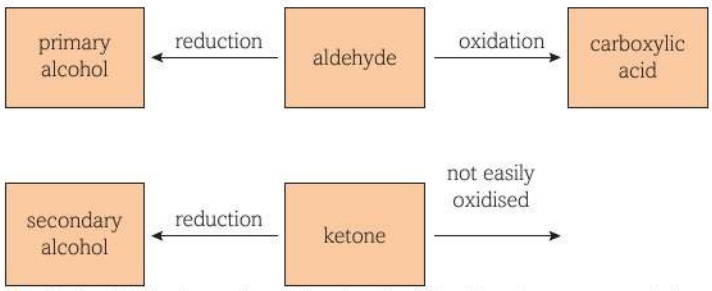
SPECIFICATION REFERENCE
15.8(i) 15.8(ii) 15.8(v)

## LEARNING OBJECTIVES

- Understand the reactions of carbonyl compounds with:
  (i) Fehling's or Benedict's solution, Tollens' reagent and acidified dichromate(VI) ions
  (ii) lithium tetrahydridoaluminate(III) (lithium aluminium hydride) in dry ether (ethoxyethane)
  (iii) iodine in the presence of alkali (the iodoform test).

## WHICH REACTIONS OCCUR?

You may remember from **Topic 10 (Book 1: IAS)** that carbonyl compounds (formed by the oxidation of alcohols) differ in their ease of oxidation – aldehydes (but not ketones) are easily oxidised to carboxylic acids. In this section, we will look at these reactions in more detail, and also consider the reduction reactions of carbonyl compounds. **Fig A** summarises these reactions.




▲ **fig A** Aldehydes can be reduced and oxidised, but ketones can only be reduced.

## REDUCTION REACTIONS

Both aldehydes and ketones can be reduced to alcohols by the same reagent, lithium tetrahydridoaluminate. This reagent is also known as lithium aluminium hydride, and its formula is $LiAlH_4$. The reduction reactions are carried out with both the carbonyl compound and reducing agent dissolved in dry ether. To simplify the equations, the reducing agent is represented by [H], which is the symbol for an atom of hydrogen provided by the reducing agent in an unspecified way. Note that this is a different use of square brackets – it does not indicate a molar concentration (as in kinetics). Here are some examples of equations.

Reduction of propanal to propan-1-ol:

$$CH_3CH_2CHO + 2[H] \rightarrow CH_3CH_2CH_2OH$$

Reduction of butanone to butan-2-ol:

$$CH_3COCH_2CH_3 + 2[H] \rightarrow CH_3CH(OH)CH_2CH_3$$

In all of these reduction reactions, a hydrogen atom becomes attached to each of the atoms in the carbonyl group.

### LEARNING TIP

The reduction reactions are included to show the conversion of aldehydes and ketones to alcohols.

The oxidation reactions are included to show how to distinguish between aldehydes and ketones.

## OXIDATION REACTIONS

Ketones are very difficult to oxidise, and, as far as this book is concerned, ketones cannot be oxidised. **Table A** shows the reagents that can oxidise aldehydes to carboxylic acids.

| REAGENT | COLOUR CHANGE | NOTES |
|---|---|---|
| Acidified potassium dichromate(VI) | orange solution → green solution | The colour change is due to the reduction of chromium from +6 to +3. |
| Fehling's solution | deep blue solution → red precipitate (see **fig B**) | In both of these solutions, the colour change is due to the conversion of a copper(II) complex to copper(I) oxide. It is not necessary to understand the difference between the chemical compositions of these two reagents. |
| Benedict's solution | deep blue solution → red precipitate | |
| Tollens' reagent | colourless solution → silver mirror (see **fig C**) | The colour change is due to the conversion of a silver(I) complex to metallic silver, which often sticks to the inside of the tube. The reagent is also described as ammoniacal silver nitrate. |

**table A**

### PRACTICAL SKILLS

All of these oxidation reactions can be carried out in the laboratory in a test tube, by mixing the carbonyl compound with the oxidising agent and leaving the mixture in a beaker of hot water.

Safety Note: Wear eye protection and avoid skin contact. Only heat in hot water from an electric kettle, NOT over a flame.

As for the reduction reaction equations, all of the oxidising agents in **table A** can be represented by [O]. This is the symbol for an

atom of oxygen provided by the oxidising agent in an unspecified way. Here are some examples of equations.

Oxidation of propanal to propanoic acid:

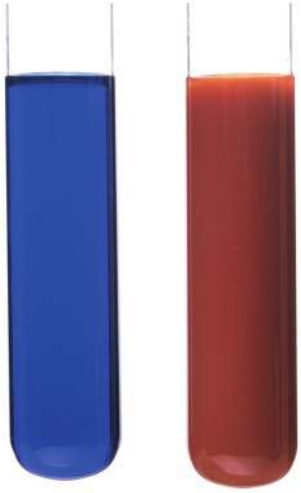
$$CH_3CH_2CHO + [O] \rightarrow CH_3CH_2COOH$$

Oxidation of butanal to butanoic acid:

$$CH_3CH_2CH_2CHO + [O] \rightarrow CH_3CH_2CH_2COOH$$

## DISTINGUISHING BETWEEN ALDEHYDES AND KETONES

The oxidation reactions using Fehling's solution, Benedict's solution and Tollens' reagent are often used to distinguish between aldehydes and ketones. A positive result, as described in **table A**, indicates the presence of an aldehyde. A negative result indicates the absence of an aldehyde. If it is also known that the substance being tested is a carbonyl compound, then the negative result can be taken as confirmation of the presence of a ketone.



▲ **fig B** Colour change of Fehling's solution from a blue solution to a red precipitate in the test for an aldehyde.



▲ **fig C** Colour change of Tollens' reagent from a colourless solution to a silver mirror in the test for an aldehyde.

## REACTIONS WITH IODINE

Another redox reaction of some carbonyl compounds is rather different – it used to be known as the iodoform reaction, but is now described as the triiodomethane reaction. This is because the product is $CHI_3$, a yellow insoluble solid.

If a carbonyl compound is added to an alkaline solution of iodine and the mixture warmed and then cooled, a pale yellow precipitate sometimes forms. This result is a positive test for carbonyl compounds containing the $CH_3CO$ group. This group is found in only one aldehyde (ethanal) and in all methyl ketones (those with the carbonyl group as the second carbon in the chain). This means that the reaction occurs with propanone, butanone and pentan-2-one, but not pentan-3-one.

Because oxidising conditions are used in the reaction, some alcohols will also give positive results. **Table B** shows the results for some common alcohols.

| ALCOHOL | OXIDATION PRODUCT | RESULT |
|---|---|---|
| ethanol | $CH_3CHO$ | positive |
| propan-1-ol | $CH_3CH_2CHO$ | negative |
| propan-2-ol | $CH_3COCH_3$ | positive |
| butan-1-ol | $CH_3CH_2CH_2CHO$ | negative |
| butan-2-ol | $CH_3COCH_2CH_3$ | positive |
| 2-methylpropan-1-ol | $(CH_3)_2CHCHO$ | negative |

**table B**

### EXAM HINT

As the iodoform test is an oxidising reaction, a secondary alcohol with a terminal methyl group (e.g. butan-2-ol) will also give a positive result.

### CHECKPOINT

**1.** What are the names of the organic products of these reactions?

(a) The reduction of $CH_3CH_2CH_2CHO$.

(b) The oxidation of $CH_3CH_2CH_2CH_2CHO$.

SKILLS REASONING

**2.** What colour changes would be observed in these tests?

(a) Heating a mixture of propanone and acidified potassium dichromate(VI).

(b) Heating a mixture of ethanal and Tollens' reagent.

# 15B 3 NUCLEOPHILIC ADDITION REACTIONS

SPECIFICATION REFERENCE
15.8(iii) 15.8(iv)
15.5 PART

## LEARNING OBJECTIVES

- Understand the reactions of carbonyl compounds with:
  (i) HCN, in the presence of KCN, as a nucleophilic addition reaction, using curly arrows, relevant lone pairs, dipoles and evidence of optical activity to show the mechanism
  (ii) 2,4-dinitrophenylhydrazine (2,4-DNPH), as a qualitative test for the presence of a carbonyl group and to identify a carbonyl compound given data for the melting temperatures of derivatives.
- Be able to use data on optical activity of reactants and products as evidence for addition to carbonyl compounds.

## REACTION WITH HYDROGEN CYANIDE

Both aldehydes and ketones react with hydrogen cyanide. The reaction is carried out in an aqueous alkaline solution containing potassium cyanide (KCN). The equations and mechanisms are the same for both aldehydes and ketones. Here are some examples of equations.

Addition to propanal:

$CH_3CH_2CHO + HCN \rightarrow CH_3CH_2CH(OH)CN$

Addition to butanone:

$CH_3COCH_2CH_3 + HCN \rightarrow CH_3C(CN)(OH)CH_2CH_3$

These addition reactions involve a hydrogen atom attaching to the oxygen atom of the carbonyl group and a cyanide group attaching to the carbon atom of the carbonyl group.

These products contain a carbon atom joined to the functional groups OH and CN. They are named differently from other organic compounds we have met so far. According to IUPAC rules, the OH group should be shown as a prefix and the CN group should be shown as a suffix. When this happens, the OH group is shown as *hydroxy-* and not as *-ol*, and the CN group is shown as the ending *-nitrile*. In general, these products are described as hydroxynitriles.

Applying these naming rules:

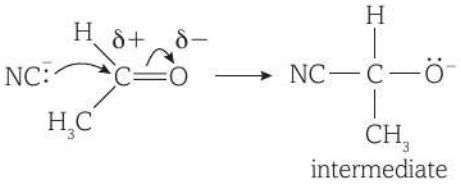
$CH_3CH_2CH(OH)CN$ is 2-hydroxybutanenitrile

$CH_3C(CN)(OH)CH_2CH_3$ is 2-hydroxy-2-methylbutanenitrile

### THE NUCLEOPHILIC ADDITION MECHANISM

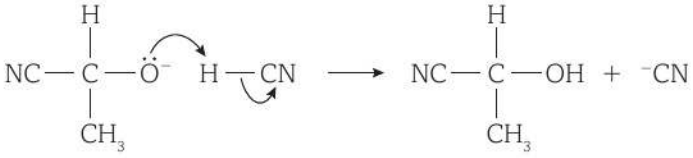
**Nucleophilic addition** is a two-step mechanism, illustrated using ethanal as an example.

Step 1 involves the nucleophilic attack by a cyanide ion on the carbon atom of the carbonyl compound.




intermediate

Step 2 involves the reaction between the intermediate and a hydrogen cyanide molecule.



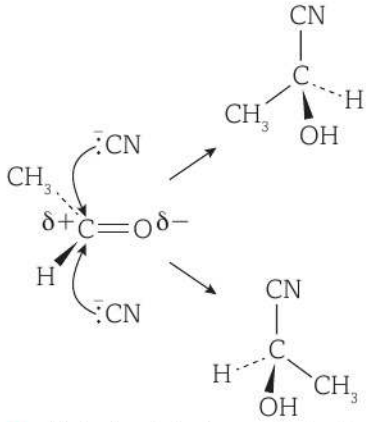

### LEARNING TIP

Remember that when a cyanide group becomes part of an organic compound, the number of carbon atoms increases by one, which affects the name. So, for example, propanal becomes 2-hydroxybutanenitrile.

### USING OPTICAL ACTIVITY AS EVIDENCE

A consideration of optical activity can be used as evidence for this mechanism. The arrangement of the two atoms or groups joined to C=O is planar, which means that in Step 1 there is an equal chance of the cyanide ion attacking from each side of the plane. So, although the product of this reaction contains an asymmetric carbon atom (a chiral centre), there are equal amounts of both dextrorotatory and laevorotatory enantiomers formed, giving a product mixture with no optical activity.

Using ethanal as the example, the product is 2-hydroxypropanenitrile. **Fig A** shows the formation of a racemic mixture containing equal numbers of both enantiomers.

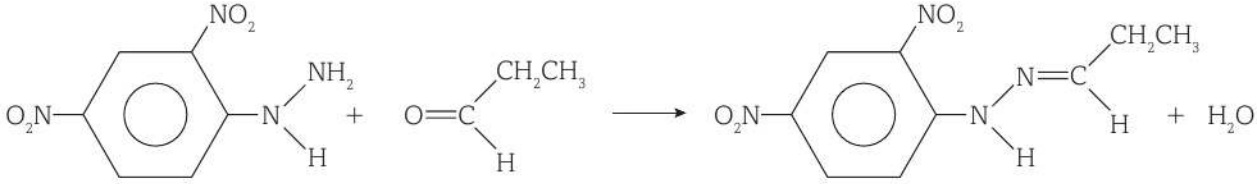



▲ **fig A** Attack by the cyanide ion from the top leads to the product shown higher up, while attack from the bottom leads to the product shown lower down. Each ethanal molecule is attacked by only one of these cyanide ions.

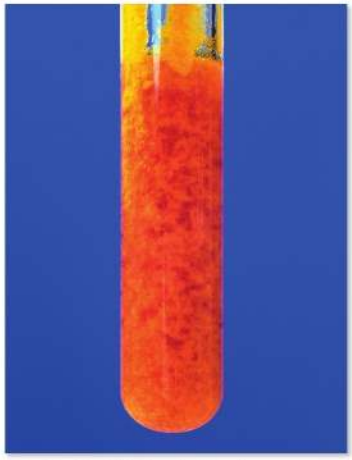
The cyanide ion can attack from above or below, producing equal amounts of both enantiomers.

## REACTION WITH 2,4-DINITROPHENYLHYDRAZINE

This compound is sometimes known as Brady's reagent, and its name is often abbreviated as 2,4-DNPH. You do not need to write equations or mechanisms for any reactions involving this reagent. However, it is useful to show its structure, and the structure of a product of its reaction with a carbonyl compound. **Fig B** shows its reaction with propanal.




▲ **fig B** The reaction between 2,4-dinitrophenylhydrazine and propanal.



▲ **fig C** This is the bright orange precipitate formed when a carbonyl compound reacts with 2, 4-dinitrophenylhydrazine.

The reagent reacts with most carbonyl compounds to form a brightly coloured orange solid (**fig C**). This means that the appearance of an orange precipitate is a positive result that indicates the presence of a carbonyl compound.

### MELTING TEMPERATURES OF DERIVATIVES

These reactions with 2,4-dinitrophenylhydrazine give a visual indication of a positive result for the presence of a carbonyl compound. They can also be used to identify individual carbonyl compounds. This is because the compounds formed, known as **derivatives** (because they are derived from particular carbonyl compounds), can be filtered, purified and dried, and their melting temperatures measured. These derivatives have the ending *-one*, so the derivative of butanal is butanal 2,4-dinitrophenylhydrazone.

The identity of the original carbonyl compound can often be confirmed by comparing the experimental values of melting temperatures of these derivatives with data book values. For example, the three carbonyl compounds pentanal, pentan-2-one and pentan-3-one all have the same boiling temperatures and all form coloured precipitates with 2,4-dinitrophenylhydrazine. **Table A** shows the boiling temperatures of the carbonyl compounds and the melting temperatures of their derivatives.

| CARBONYL COMPOUND | BOILING TEMPERATURE/°C | MELTING TEMPERATURE OF 2,4-DINITROPHENYLHYDRAZONE DERIVATIVE/°C |
|---|---|---|
| pentanal | 102 | 104 |
| pentan-2-one | 102 | 144 |
| pentan-3-one | 102 | 156 |

**table A**

You can see that the original carbonyl compound can now be identified by matching the experimental melting temperature of its derivative to a value in the table.

## CHECKPOINT

1. What are the names of the organic products of these reactions?
   (a) methanal + hydrogen cyanide →
   (b) propanone + hydrogen cyanide →
2. Compound X reacts with 2,4-dinitrophenylhydrazine and with iodine in aqueous alkali in separate tests. In both cases a coloured precipitate forms. Explain which of these could be compound X:
   propanal, propan-1-ol, propan-2-ol, propanone

## SUBJECT VOCABULARY

**nucleophilic addition** a type of mechanism in which a molecule containing two atoms or groups is added across a polar double bond (usually C=O) and the attacking species in the first step is a nucleophile
**derivatives** compounds formed from other compounds, especially when the properties of the derivatives can be used to identify the original compound